The GEOMETRY of shapes

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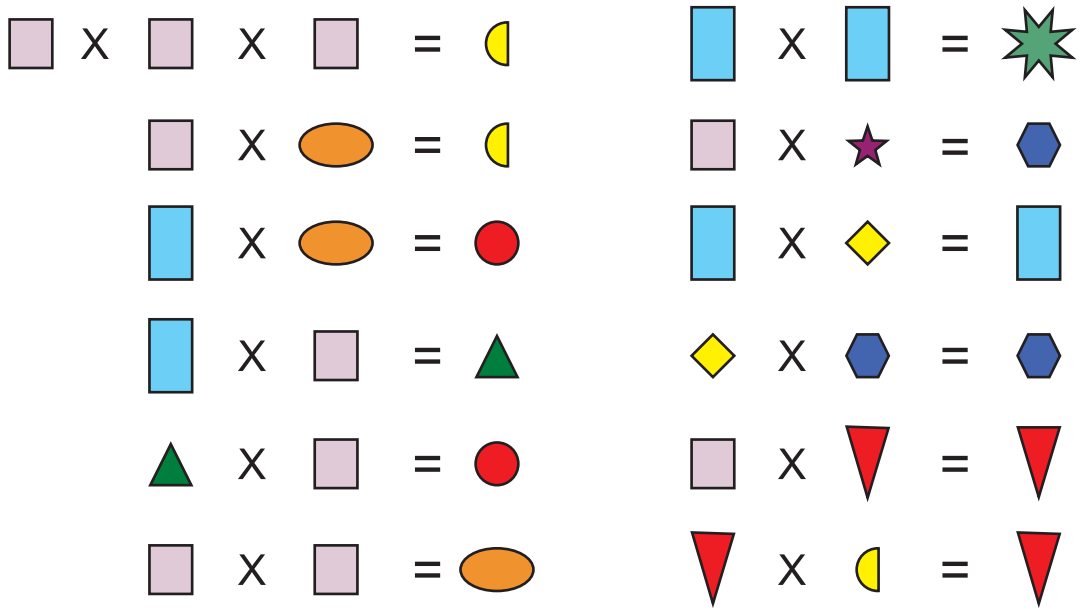
Activity 1: Flatland movie

This movie is a perfect start for a geometry introductory class with age groups 10-11. The characters in the form of square, equilateral triangle, pentagon, hexagon and the "circle priest" move around in 2-dimensional space. It is a movie where little "hex" asks a lot of questions on the three dimensional existence and how circle priests deal with it! The story is based on the science fiction Flatland: A Romance of Many Dimensions written by Edwin A. Abbott. This movie is available online.

Activity 2: Worksheet: Shape times Shape

The shapes stand for eleven of the numbers from 0 to 12. Each shape is a different number.
Can you work out what they are from the multiplications below?

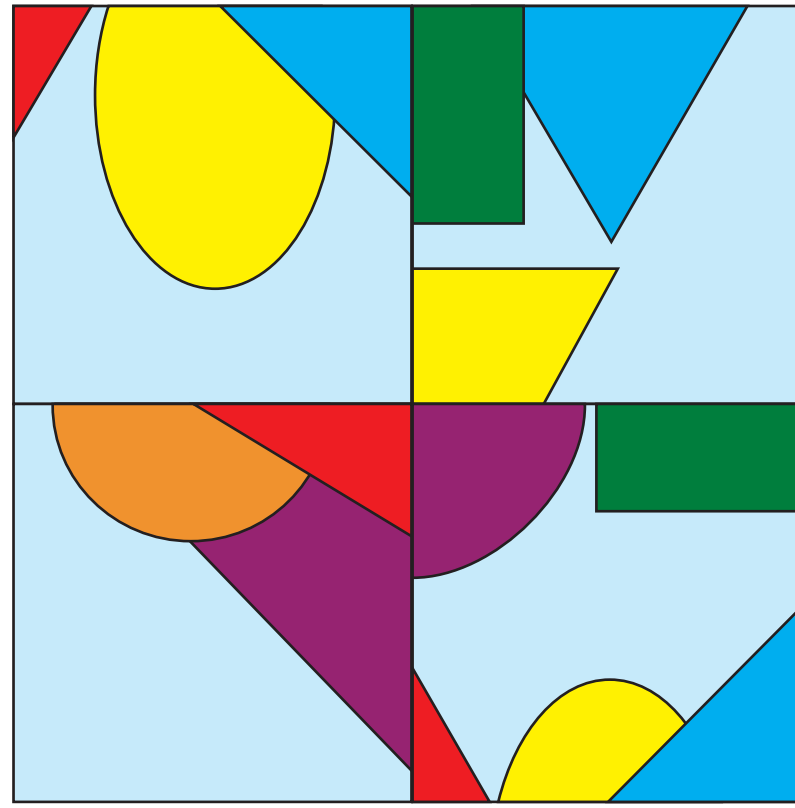


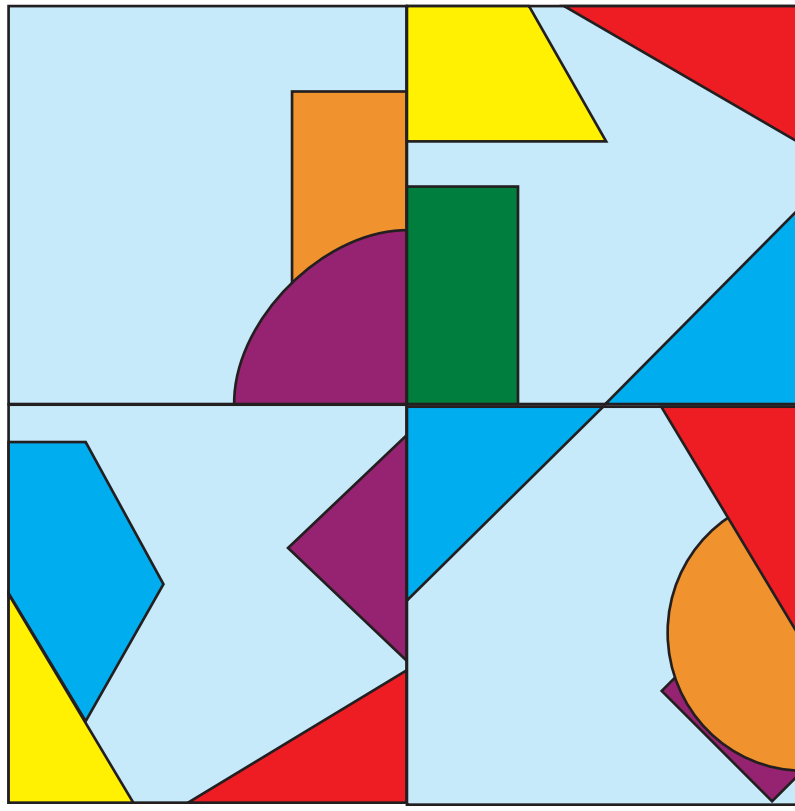
Square =	8 pointed star =	Semi circle =	Rectangle =	Oval =	Five pointed star =
Hexagon =	Circle =	Diamond =	Equilateral triangle =	Isosceles triangle =	

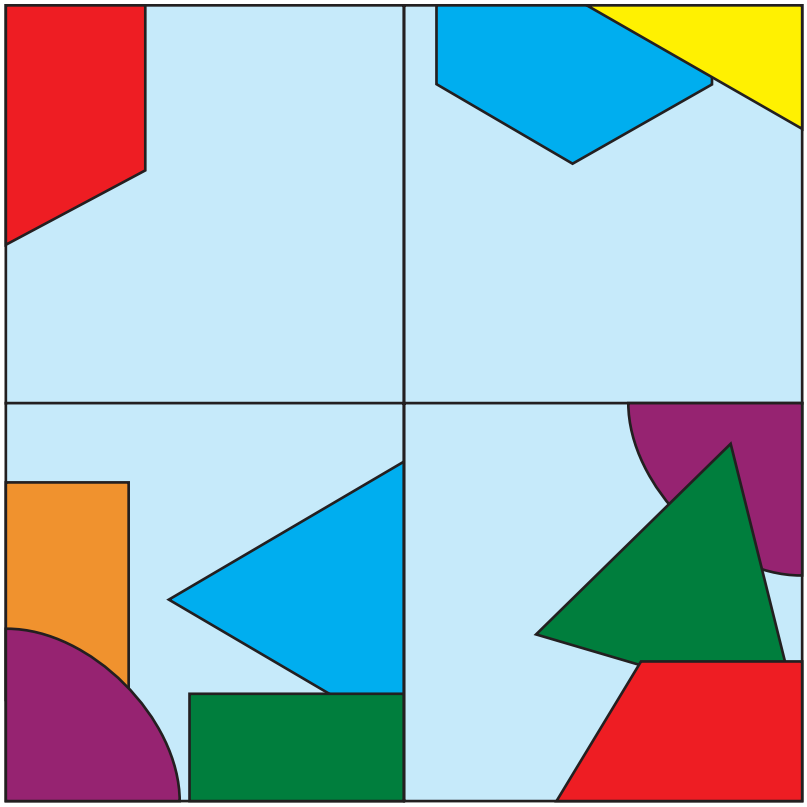
Activity 3: Puzzle – Jig Shapes

Group of 4

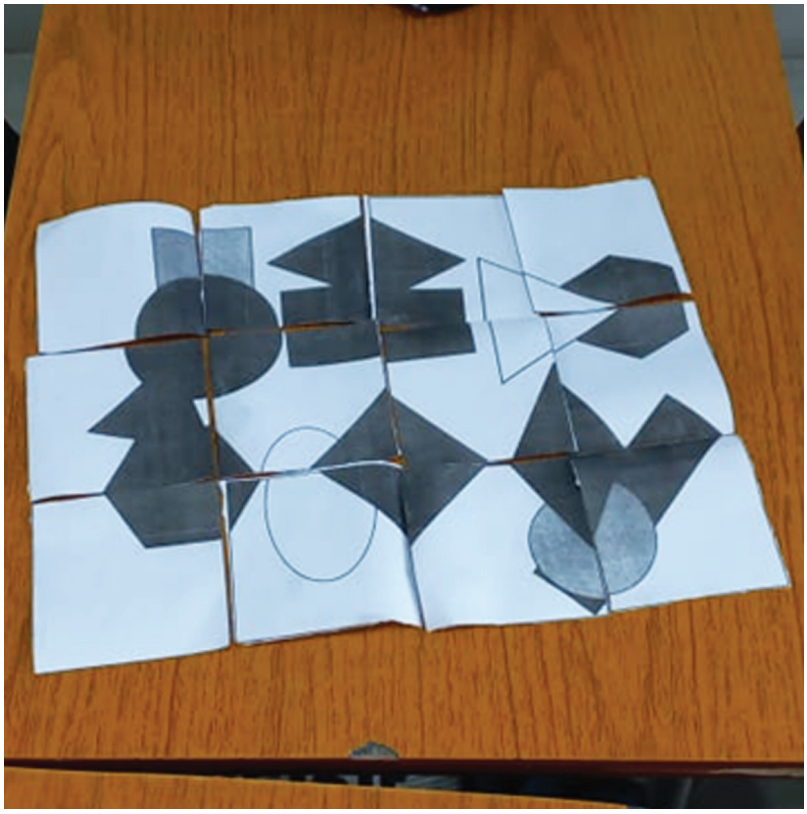
- Print out the sheets. The sheets will need to be cut into twelve separate cards.
- Share all the cards among the group members.
- Students are not supposed to show their cards to the group members. They need to describe their cards to others by using mathematical terminologies. In turn everyone in the group will describe their cards and accordingly the rest need to find the matching pieces from their cards and put it together to make a one whole picture.
- The group activity should develop listening skills, team work, mathematical language about shape and position.







By the end of the puzzle, it will look like:



Activity 4: Estimating angles

This can be played among two teams. The teacher will require a computer and internet. One team member comes and tries to make the estimated angle shown on the screen. The closer they get to the target angle the more points they will score.

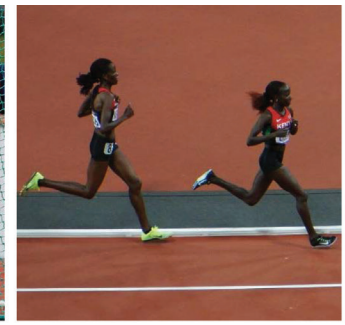
The scoring system:

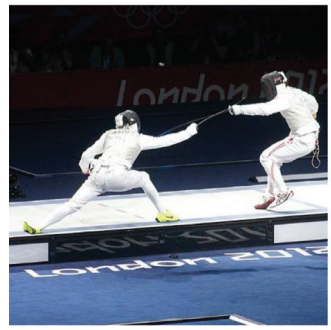
Error	Points
0-5°	10
6-10°	5
11-15°	2

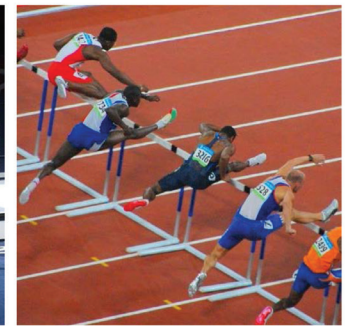
Activity 5: Olympic Turns

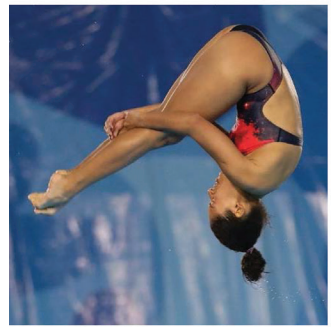
Group of 4 to 5
Different sports involve angles in different ways. Gymnastics involves making the body twist, turn and spin to accomplish different feats. Ball games involve choosing the angle at which to pass the ball to another player or to aim for the goal.

Here are photos of some Olympic sports that involve turns and angles in different ways. Choose one of these photos to investigate and see what angles you can find. You can draw the straight lines on pictures and then estimate or measure the angles that are made by these lines.









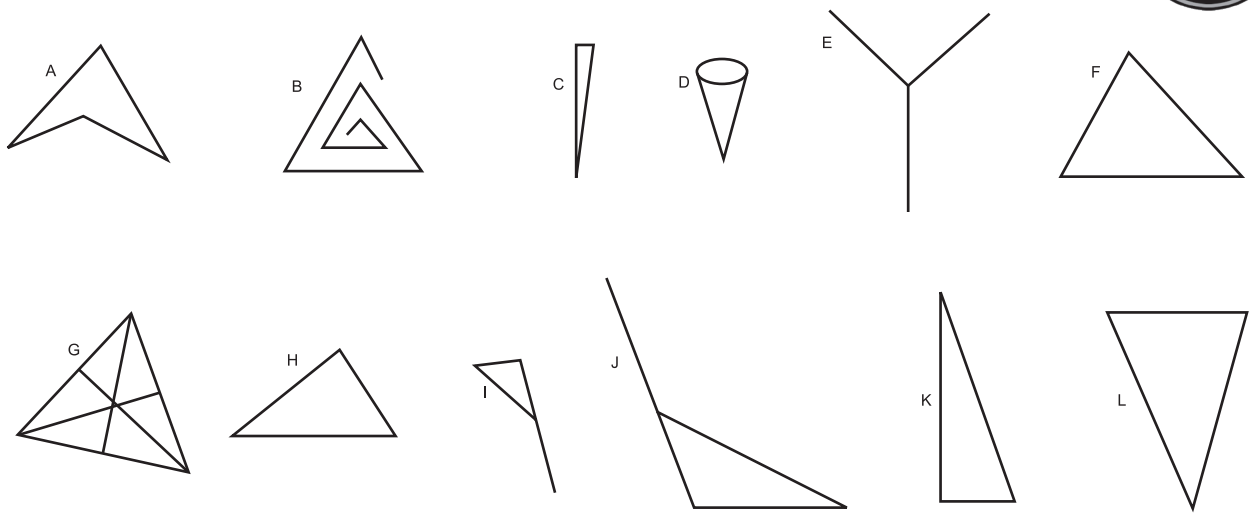
Key questions
1. Which angles are important? Why?
2. What would change if that angle was bigger/smaller? Why has the athlete chosen to position their body at that angle?
3. Can you estimate that angle? Can you measure it?

Possible support
Look at a video clip. Are there any angles involved in the sport that are important? Are there turns involved? How much does the athlete turn? How do the angles at which they hold their arms, their legs or pieces of equipment affect their success?

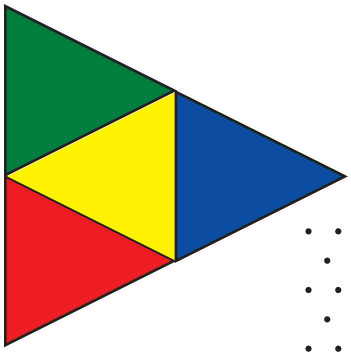
Olympic Website: https://olympics.com/en/sports/

Activity 6: Triangles

1. Which of these is a TRIANGLE? And why? And why is it not?



2. A triangle which has all the sides equal is called an EQUILATERAL TRIANGLE. An equilateral triangle can be divided into four (equal) smaller **equilateral** triangles.



Is it possible to dissect a larger equilateral triangle into any number of smaller equilateral triangles of the same size? Can you find a pattern?

Use the triangular dotted paper below to draw different sizes of equilateral triangles and divide it into smaller equilateral triangles to find a pattern.

Activity 7: Islamic Art

This is a part of a monument in Fatehpur Sikri. We can see some beautiful geometric patterns here!

Image

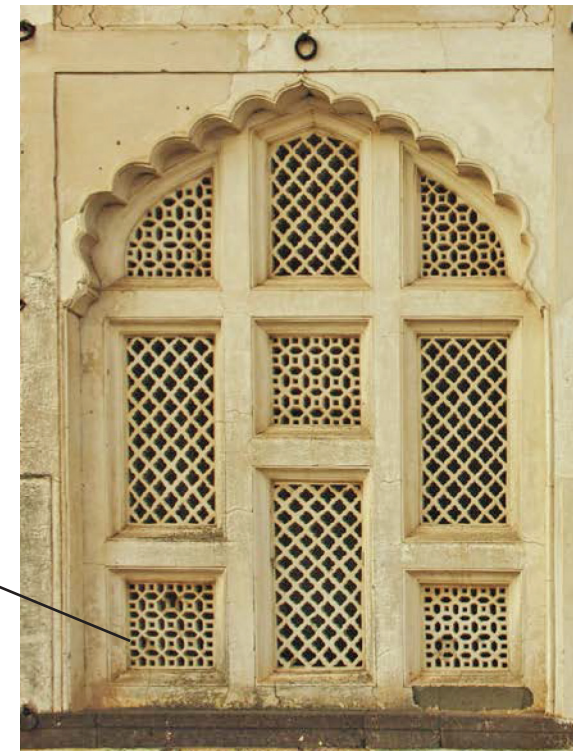
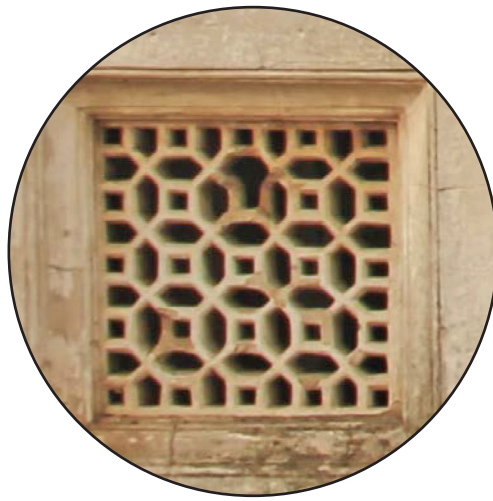
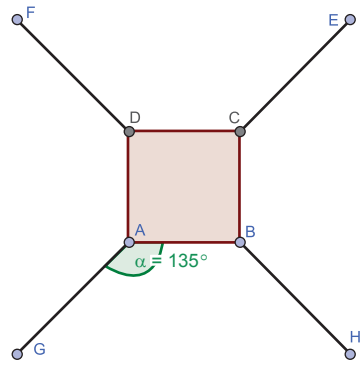


Photo courtesy: pexels.com

Zoomed version of Image will look like:



If we look at one unit of it:



Let us try to recreate the above image using pencil, straightedge and compass.

Hint 1: Construct a square (all 90° should be constructed using compass)
Hint 2: How do we make 135° with a straightedge and compass?

After creating one unit, you can continue the pattern and recreate this beautiful art on paper or maybe do a glass painting?!

References

- Website: https://nrich.maths.org/
- Rishi Valley Worksheet
- Azim Premji University – Lecture on Islamic Art and Maths

Krittika Hazra is a middle school maths educator in The Future Foundation School, Kolkata. She takes keen interest in designing and restructuring the curriculum. She passionately involves with students to understand their mindset about the subject and tries to develop appropriate content. She can be reached at <krittika.hazra7@gmail.com>.